



Module 10

Neuro Symbolic AI & Architectures

*"Can a machine that recognizes faces also understand WHY?
Bridging perception and logic."*

Where We Are in the Journey

- ✓ Module 02: Biological Neuron
- ✓ Module 03: Neural Coding
- ✓ Module 04: Synaptic Plasticity
- ✓ Module 05: Vision & Perception
- ✓ Module 06: Memory Systems
- ✓ Module 07: Attention & Consciousness
- ✓ Module 08: Reinforcement Learning
- ✓ Module 09: Language & Cognition
- **Module 10: Neuro Symbolic AI & Architectures**

"Can You Solve This?"

Five houses. Five nationalities. Five pets.

The Brit lives in the red house.

The Swede has a dog.

The Dane drinks tea...

How did you solve that?

What parts of your brain were doing what?

Neural recognition (reading the clues) + **Symbolic reasoning** (applying the constraints)
= YOUR BRAIN doing neuro symbolic AI



Intelligence Has Two Faces

NEURAL INTELLIGENCE

Pattern Recognition

Fast, automatic, fuzzy,
parallel

Recognizes a face in 100ms

Can't explain HOW it did it

SYMBOLIC INTELLIGENCE

Logical Reasoning

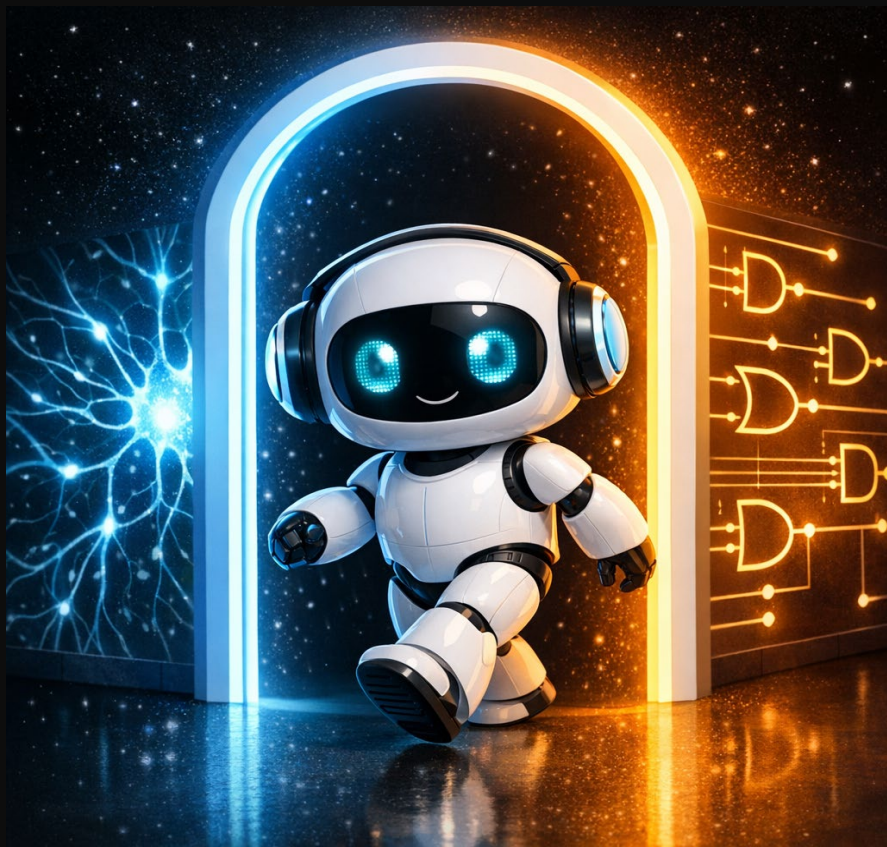
Slow, deliberate, precise,
serial

Applies rule to new cases

Can explain every step



"The brain has both. Most AI has only one."



PART 1

The Two Minds of AI

Neural Networks: Amazing Pattern Machines

- **Image recognition:** better than humans on ImageNet
- **Game playing:** superhuman on chess, Go, Atari
- **Language:** GPT 4 writes poetry, code, essays

The secret: millions of examples + billions of parameters
= very compressed pattern memory

**"A neural net doesn't know rules. It knows patterns so
deeply
compressed they LOOK like rules."**



Neural Networks Have a Dirty Secret

1. **The adversarial example:** add invisible noise to a photo
→ a panda becomes a gibbon with 99% confidence
2. **The spurious correlation:** model classifies “wolf” correctly because there’s snow in every wolf photo
3. **The black box:** you cannot ask WHY it made that decision
4. **The data hunger:** needs millions of examples for each new task (humans need one)

"If you can fool it with a sticker, does it really understand?"



Symbolic AI: The Logic Machine

Symbolic AI represents knowledge as rules, facts, and relationships.

Classic wins:

Deep Blue (chess, 1997): evaluated 200 million positions/second using explicit rules

MYCIN (1970s): diagnoses bacterial infections using 600 IF THEN rules

Wolfram Language: 100,000+ built in mathematical and logical functions

Advantage: You can ask it WHY. Every step is traceable and explainable.



The Symbol Grounding Problem

How does the symbol “RED” connect to the actual experience of red?

How do you write a rule for “a natural looking smile”?

The brittleness problem: any situation the rules don't cover → failure

Famous example: Robot that could navigate a clean lab perfectly confused by a paper bag on the floor (not in the rules)

**"You cannot write enough rules to cover the real world.
There are always exceptions to exceptions."**



⚠️ “Symbolic AI Was Proven to Fail”

NO. Symbolic AI was proven to have **LIMITS** not to fail.

The 1980s AI Winter happened because researchers **ONLY** used symbolic AI for problems that needed perception.

Those limits are exactly what neural nets fill in.

Modern view: neural nets for perception + symbolic systems for reasoning = best of both worlds

"Throwing away symbolic AI because it can't see is like throwing away calculators because they can't drive."



Neuro Symbolic = Neural + Symbolic

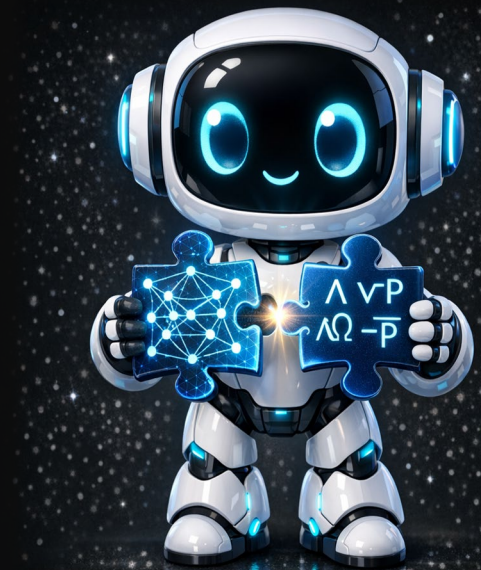
Neural: perceives the world (images, sound, text, sensor data)

Symbolic: reasons about the world (rules, logic, knowledge graphs)

The brain does both: visual cortex (neural, fast) + prefrontal cortex (symbolic, slow)

4 Integration Strategies:

1. **Sequential:** neural perceives → symbolic decides
2. **Guided:** symbolic rules constrain neural training
3. **Compiled:** symbolic rules distilled into neural weights
4. **Bidirectional:** constant feedback loop





PART 2

The Neuro Symbolic Brain

System 1 + System 2: Neuro Symbolic Brain

SYSTEM 1 = NEURAL

Fast, parallel, automatic,
unconscious

Brain: sensory cortices,
basal ganglia, cerebellum

Like: face recognition,
driving familiar roads

SYSTEM 2 = SYMBOLIC

Slow, serial, deliberate,
conscious

Brain: prefrontal cortex,
working memory

Like: solving a logic puzzle,
planning a new trip



**"The brain has been doing neuro symbolic AI for 300
million years.**

We're just figuring out how to copy it."

From Examples to Rules

You see 10 chairs: different shapes, colors, sizes.

You extract the **CONCEPT** of “chairness.”

Now you recognize chairs you’ve never seen before even chairs that break every rule.

This is concept learning: inferring abstract structure from concrete examples.

One shot learning: children learn new words in one exposure. Current AI needs thousands.

**"The brain does not memorize examples.
It extracts the DNA of a concept."**



Your Brain Is a Knowledge Graph

The brain's 86 billion neurons form structured circuits that represent relationships.

Example: "Dog" → connected to "animal," "fur," "loyal," "bark," "danger," "playful"

When you damage specific brain regions, specific semantic categories disappear:

Patient S.B.: could identify tools but not living things.
Brain wiring = semantic organization.

AI equivalent: Google Knowledge Graph, Wikidata, ontologies



**"The brain does not store facts in isolation.
It stores them in a web of relationships."**



PART 3

Neuro Symbolic Architectures in AI

Global Workspace Theory

Bernard Baars (1988): consciousness arises from a GLOBAL WORKSPACE a central broadcast medium.

Many specialist modules (vision, hearing, motor, memory) compete for access.

Winner broadcasts to ALL other modules simultaneously.

This is what a neuro symbolic architecture needs:

- Neural specialist modules (perception, language)
- Shared symbolic working memory (the global workspace)
- Selective broadcasting = attention mechanism

"Your context window is a crude version of global workspace.

The brain's workspace is much richer."



Neuro Symbolic AI in the Wild

AlphaFold (DeepMind)

Neural nets + biophysics constraints.

Predicted 200M protein structures.

Nobel Prize 2024.

Google Search

Neural ranking + Knowledge Graph.

“Who is the CEO of Apple?” → neural finds pages, KG gives the exact answer.

DreamCoder (MIT)

Learns programs from examples.

Combines neural pattern recognition with symbolic program synthesis.

Wolfram Alpha

Neural natural language understanding + symbolic mathematics engine.



Wolfram Language: Symbolic at Scale

Wolfram Language is built around symbolic computation

- Every expression is a symbolic tree
- Mathematics, logic, data, visualization, AI all unified
- 100,000+ built in functions representing decades of curated knowledge
- Natural language input → symbolic query → precise answer

For this class: Wolfram Language handles the reasoning/logic layer;
Python handles the neural/ML layer



"Wolfram Language is to symbolic AI what Python is to neural AI."

Building the Bridge

PYTHON (Neural Layer)

Keras/TensorFlow classifier
Extracts features from images
Outputs symbolic labels
(e.g., "red_house",
"british_man")

WOLFRAM (Symbolic Layer)

Takes labels as facts
Applies logical constraints
Uses Solve[] or Reduce[]
to find the solution



Integration: Python outputs → Wolfram inputs → logical answer

**"This is exactly how the brain works:
visual cortex (neural) → prefrontal cortex (symbolic)"**

? Knowledge Check

1. A self driving car uses a neural net to detect pedestrians, but uses a rule based system to decide whether to stop. Is this neuro symbolic? Which layer does what?
2. Why can't a pure deep learning system reliably solve the Zebra puzzle? What's missing?
3. Global Workspace Theory says consciousness = global broadcast. What does this mean for an AI that needs to combine vision, language, and reasoning?





AGI

PART 4

The Road to AGI

Is Deep Learning Enough for AGI?

YES Scaling Camp

Altman, Sutskever

"GPT 5 will figure it out. We just need more compute and data."

Evidence: GPT 4 passed the bar exam, doctors, coding interviews

NO Hybrid Camp

Bengio, Marcus, LeCun

"Scaling a pattern matcher cannot reach causal, physical reasoning."

Evidence: LLMs fail on novel reasoning, multi step planning

"This is the most important debate in AI right now.

You are learning the vocabulary to participate in it."



Yann LeCun's Alternative: JEPA

LeCun argues: we don't need symbolic rules IF we have the right architecture for learning world models.

JEPA (Joint Embedding Predictive Architecture):
Learn representations that capture causal structure

Key ideas:

- Self supervised learning from video
- Hierarchical world models
- No language needed for intelligence

Difference from LLMs: models abstract reality directly, not language descriptions of reality



"LeCun believes the path to AGI goes through robots learning from the physical world, not text on the internet."

What Neuro Symbolic AI Still Can't Do

- **Symbol grounding:** connecting abstract symbols to real world perception at scale
- **Causal reasoning:** “X causes Y” vs. “X correlates with Y”
- **Compositionality:** combining known concepts in novel ways (a child does this effortlessly)
- **Common sense physics:** “if you push a glass off a table, it falls”
- **Long horizon planning:** 100 step sequences with changing goals

"These are YOUR research opportunities. This field is wide open."



Neuro Symbolic AI in Industry

IBM Research

Neuro symbolic reasoning
for enterprise AI.

Role: Research Scientist

DeepMind

AlphaFold, Gato, and
successor architectures.

Role: ML Engineer

MIT CSAIL

DreamCoder, ARC
benchmark research.

Role: Research Assistant /
PhD

Startups

Rebellion Research,
Symbolica AI, Numenta.

Role: AI Engineer



**"This is the frontier. The engineers who master neuro symbolic AI
will define the next decade of AI."**

What You Learned in Module 10

Neural AI

Pattern recognition, fast, fuzzy, data hungry

Symbolic AI

Logical reasoning, slow, precise, brittle at perception

Neuro Symbolic AI

Hybrid architecture perception + reasoning

System 1/2 Architecture

Brain's biological neuro symbolic design

Global Workspace Theory

Shared broadcast workspace integrating specialist modules

Concept Learning

Extracting abstract structure from concrete examples

Next: Motor Control & Robotics



Module 10 asked: **How does the brain REASON?**

Module 11 asks: **How does the brain MOVE?**

Embodied Intelligence: intelligence that lives in a body

The Cerebellum: predictive motor control at millisecond speed

Neuromorphic robotics: low latency control inspired by the brain

Preview: Simulating a robotic arm with neural adaptation



PART 5

Building It: Lab 10 Preview

Lab 10: Neuro Symbolic Puzzle Solver

Task: Solve an Einstein style logic puzzle using both Python (neural perception layer) and Wolfram Language (symbolic reasoning layer)

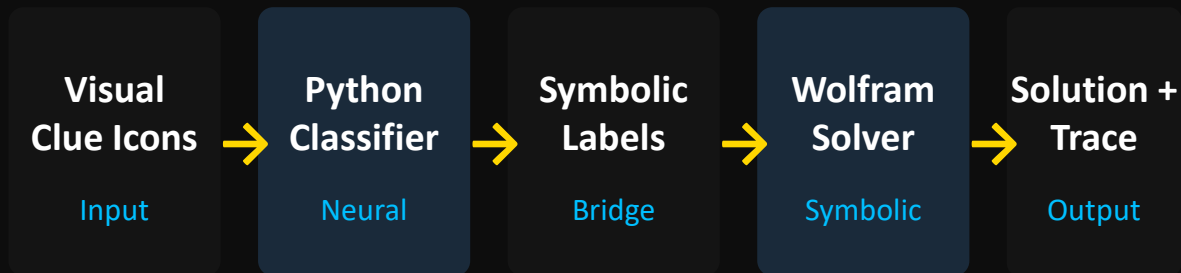
Architecture:

- 1. Python component:** A simple neural classifier reads visual clue icons → outputs symbolic labels
- 2. Wolfram component:** Takes those labels as constraints → uses symbolic solver to find the unique solution

Output: The correct solution + a traceable reasoning chain



How the Two Systems Work Together



Neural (Python) | Symbolic Bridge | Symbolic Reasoner (Wolfram)

**"This is a toy version of how Google Search works:
neural
understanding of your question + symbolic knowledge
graph answer."**

